ISM6359 Data Mining

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DATA MINING PROJECT – PATIENT HEALTH RECORD

Presented by Tejashwini Saravanan

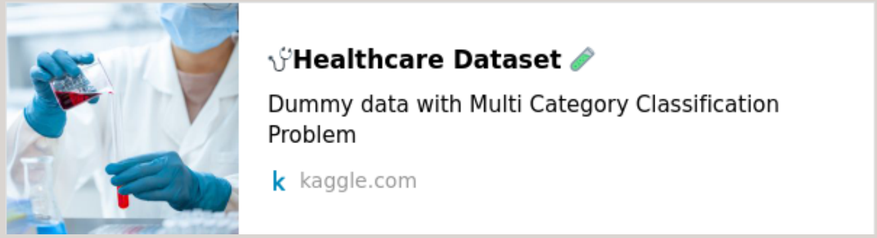
“The Prediction of Patient Healthcare Records"

- The reason I selected Orange software as my tool is because it offers a user-friendly interface coupled with powerful data processing functions.
- This combination allows for efficient analysis and manipulation of synthetic patient healthcare records.
- Specifically, I aim to use Orange to identify patterns and predict admission types for patients, leveraging its intuitive interface and robust analytical capabilities to streamline this process effectively.

orange
DATA MINING

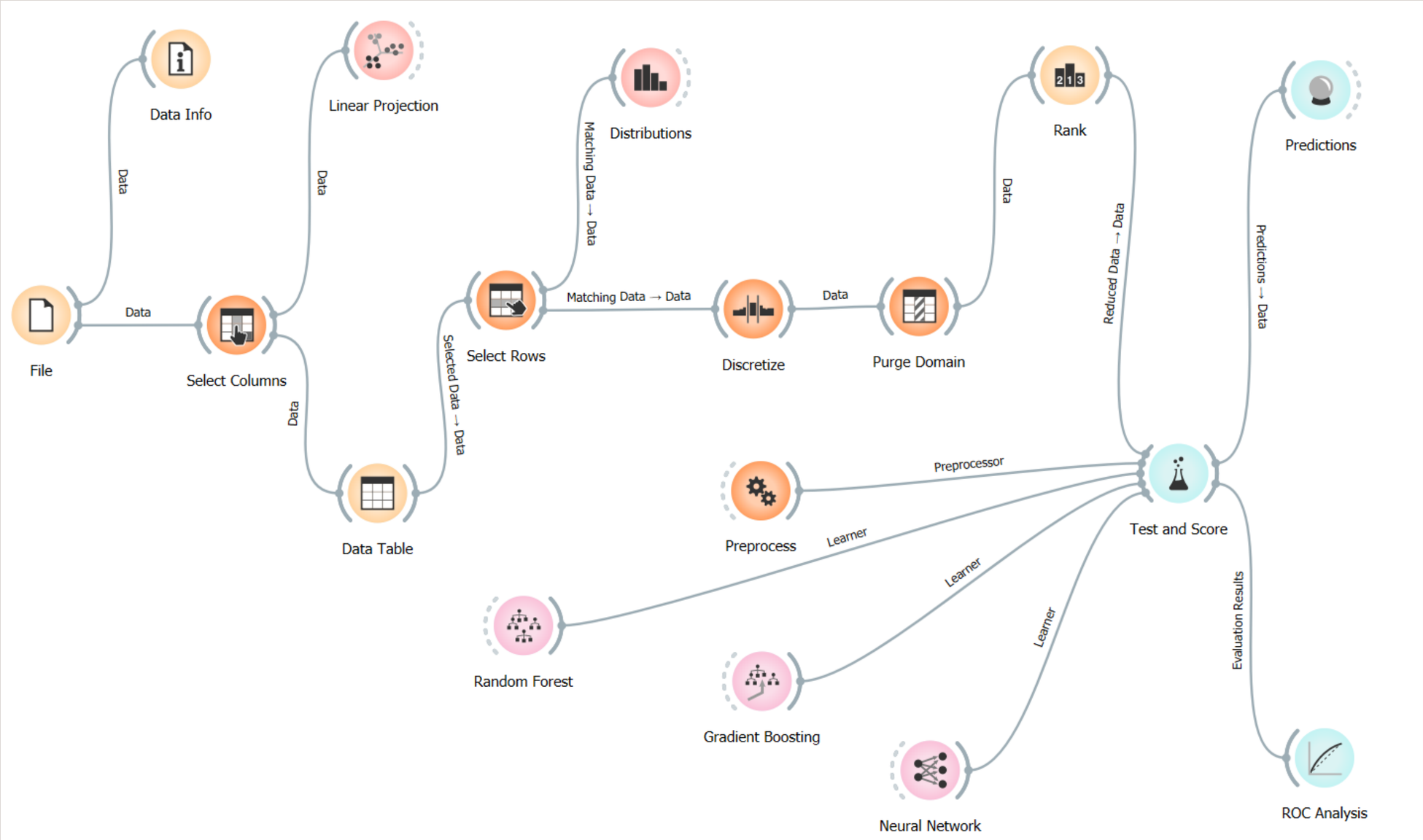
Medical record

The data utilized in this project was sourced from Kaggle



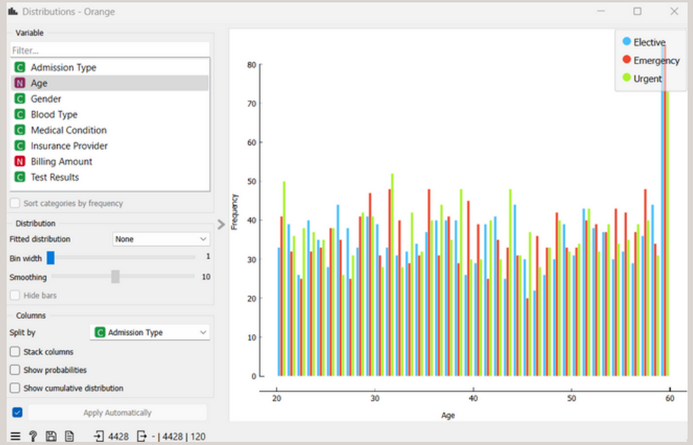
- The dataset comprises 10,000 synthetic patient healthcare records encompassing diverse attributes. These include patient demographics, encompassing age, gender.
- Medical conditions are documented within the dataset, providing insights into the health status and history of each patient. Additionally, admission details such as the room number and date of admission are recorded, shedding light on the circumstances surrounding each patient's hospitalization
- Furthermore, crucial financial aspects are captured, including the billing amount associated with each patient's stay, along with details regarding their insurance provider.

“WORK PROCESS”



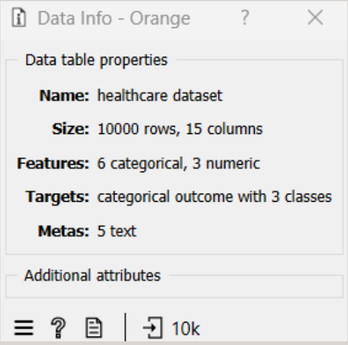
DATA TABLE

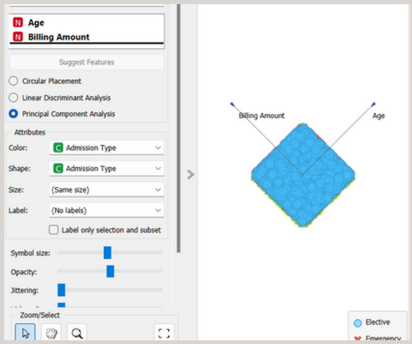
	Admission Type	Name	Hospital	Age	Gender	Blood Type	Medical Condition	Insurance Provider	Billing Amount	Test Results
1	Elective	Tiffany Ramirez	Wallace-Hamilt...	81	Female	O-	Diabetes	Medicare	37491	Inconclusive
2	Emergency	Ruben Burns	Burke, Griffin a...	35	Male	O+	Asthma	UnitedHealthcare	47304.1	Normal
3	Emergency	Chad Byrd	Walton LLC	61	Male	B-	Obesity	Medicare	36874.9	Normal
4	Urgent	Antonio Frederi...	Garcia Ltd	49	Male	B-	Asthma	Medicare	23303.3	Abnormal
5	Urgent	Mrs. Brandy Flo...	Jones, Brown a...	51	Male	O-	Arthritis	UnitedHealthcare	18086.3	Normal
6	Urgent	Patrick Parker	Boyd PLC	41	Male	AB+	Arthritis	Aetna	22522.4	Abnormal
7	Urgent	Charles Horton	Wheeler, Bryant...	82	Male	AB+	Hypertension	Cigna	39593.4	Abnormal
8	Elective	Patty Norman	Brown Inc	55	Female	O-	Arthritis	Blue Cross	13546.8	Normal
9	Elective	Ryan Hayes	Smith, Edwards ...	33	Male	A+	Diabetes	Aetna	24903	Abnormal
10	Urgent	Sharon Perez	Brown-Golden	39	Female	O-	Asthma	Blue Cross	22788.2	Normal
11	Emergency	Amy Roberts	Little-Spencer	45	Male	B-	Cancer	Aetna	40325.1	Abnormal
12	Emergency	Mrs. Caroline F...	Rose Inc	23	Female	O-	Hypertension	Medicare	6185.9	Inconclusive
13	Elective	Christina Willia...	Malone, Thomp...	85	Female	A+	Diabetes	Aetna	4835.95	Inconclusive
14	Elective	William Page	Richardson-Po...	72	Female	A+	Diabetes	Cigna	13669.4	Normal
15	Emergency	Michael Bradsh...	Castaneda-Hardy	65	Female	AB+	Cancer	Cigna	10342.8	Inconclusive
16	Emergency	Brian Dorsey	Burch-White	32	Female	O+	Arthritis	Aetna	27174.9	Inconclusive
17	Elective	Olivia Gonzalez	Cunningham an...	64	Male	AB-	Diabetes	Aetna	17395	Inconclusive
18	Elective	Teresa Caldwell	Bell, Mcknight a...	23	Male	A+	Arthritis	Medicare	45213.5	Inconclusive
19	Elective	Desiree William...	Pugh-Rogers	66	Male	O+	Obesity	UnitedHealthcare	4262.91	Inconclusive
20	Emergency	Sally Shaw	Rush, Owens an...	80	Male	O-	Arthritis	Blue Cross	16609.3	Inconclusive
21	Emergency	William Johnson	Pearson LLC	55	Female	AB+	Arthritis	Aetna	32263.6	Normal
22	Urgent	Steven Bennett	Schultz-Powers	79	Male	O-	Asthma	Blue Cross	42610.7	Abnormal
23	Emergency	Haley Li	Jordan Inc	51	Male	B-	Obesity	Medicare	16701.3	Abnormal
24	Urgent	Angela Brown	Lewis-Nelson	33	Female	B-	Diabetes	UnitedHealthcare	22331.3	Abnormal
25	Urgent	Beverly Miller	Vaughn PLC	54	Male	A-	Asthma	Cigna	41319.5	Normal
26	Urgent	Daniel Dickson	Mcknight-Mclean	26	Female	B-	Obesity	UnitedHealthcare	37766.5	Abnormal
27	Elective	Kimberly Mason	Gilbert and Sons	70	Female	B-	Obesity	Aetna	35834.4	Abnormal
28	Emergency	Francis Newman	Hess-Lowe	74	Female	AB+	Hypertension	Cigna	12680.7	Abnormal
29	Elective	Ronnie Hughes	Hawkins Ltd	56	Female	A-	Cancer	Aetna	12935.4	Abnormal
30	Emergency	Shannon Fitzpa...	Miller-Grimes	35	Female	O+	Hypertension	UnitedHealthcare	18393.4	Inconclusive
31	Emergency	Teresa Gonzalez	Sanders Ltd	37	Female	B-	Arthritis	Medicare	22356.1	Abnormal
32	Emergency	Rodney Maynard	Nixon, Evans an...	81	Male	AB-	Obesity	Blue Cross	21001.3	Normal
33	Urgent	Kevin Johnson	Hernandez, Hu...	78	Male	O-	Asthma	Medicare	9007.97	Abnormal
34	Elective	Rebecca Parsons	James and Sons	51	Male	B-	Asthma	Blue Cross	39031.4	Abnormal
35	Emergency	Linda Chavez	Winters and Sons	33	Male	A-	Cancer	Cigna	24609.4	Abnormal
36	Emergency	Jennifer Rodrig...	Duke Group	83	Female	AB-	Asthma	Cigna	5098.66	Abnormal
37	Emergency	Anna Adams	Snow Group	70	Female	A+	Obesity	Medicare	1000.18	Abnormal



- To analyse my dataset, within the tool. Firstly, I employed the "Data Info" function to gain insights into the properties of my data table.

- Next, I utilized the "Select Columns" tool to specify my target variable, which in this case was admission type, while disregarding unnecessary columns.

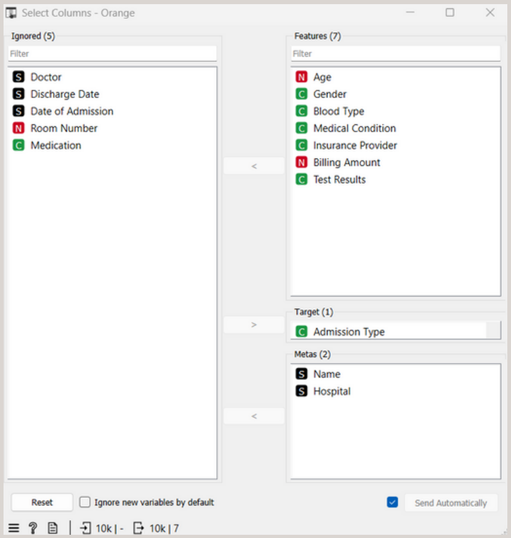




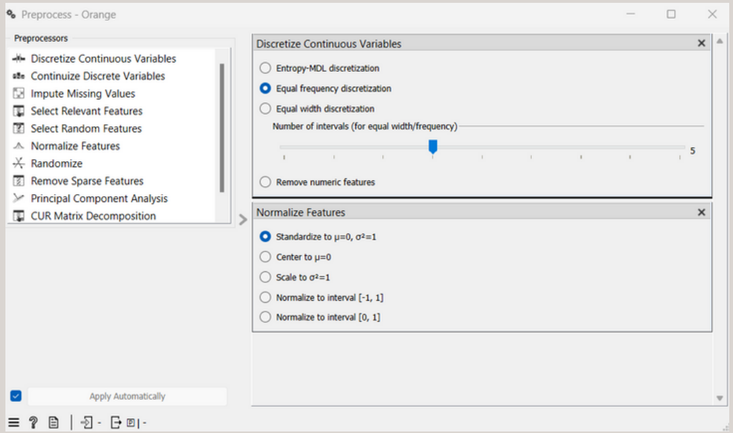
- Following this, I employed "Linear Projection" to examine the relationship between age and billing amount, utilizing Principal Component Analysis to delve deeper into attribute correlations.

- After reviewing these insights, I proceeded to filter my data using the "Select Rows" tool, applying conditions such as age between 20 to 60, billing amount exceeding 5500, and excluding cases where the insurance provider was Medicare.

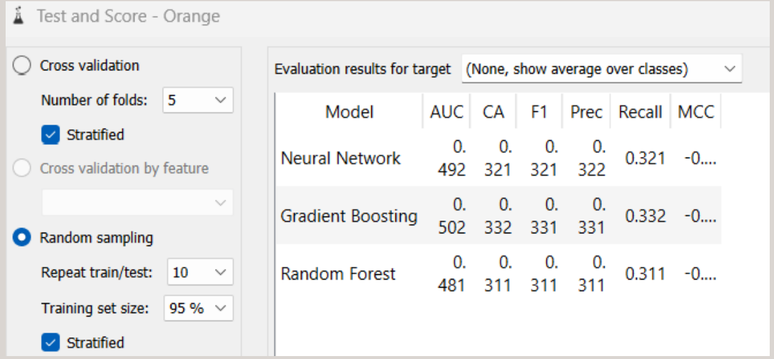
- Finally, leveraging the "Distribution" feature within the visualization tool, I generated a bar graph representing the distribution of ages within the filtered dataset.



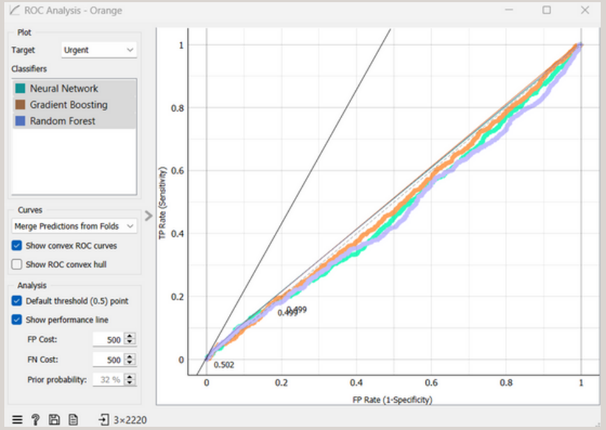
- I began by selecting specific rows to discretize and configured the desired bins for 10. This process was followed by purging the domain with all features, classes, and meta attributes enabled.
- Subsequently, I employed the RANK function to evaluate scoring methods such as information gain, gain ratio, Chi-squared, and Gini decrease, focusing on the top 4 attributes. To further enhance the data, I applied preprocessing techniques to discretize continuous variables and normalize features.

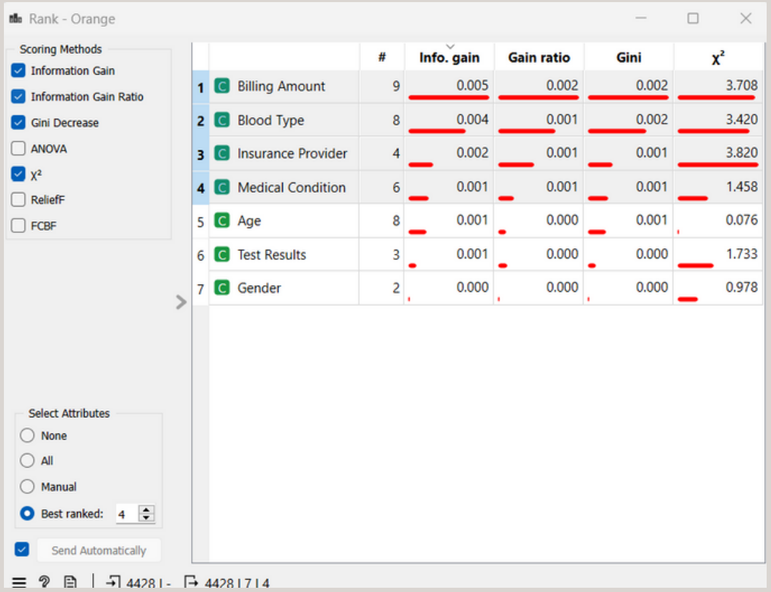


- Moving on to model selection, I employed Random Forest, Gradient Boosting, and Neural Network algorithms, connecting them to test scores for evaluation. Finally, I conducted ROC analysis with the target labeled as "urgent," examining ROC curves and associated analysis to gauge the predictive performance comprehensively.



Model	AUC	CA	F1	Prec	Recall	MCC
Neural Network	0.492	0.321	0.321	0.322	0.321	-0....
Gradient Boosting	0.502	0.332	0.331	0.331	0.332	-0....
Random Forest	0.481	0.311	0.311	0.311	0.311	-0....





		#	Info. gain	Gain ratio	Gini	χ^2
1	Billing Amount	9	0.005	0.002	0.002	3.708
2	Blood Type	8	0.004	0.001	0.002	3.420
3	Insurance Provider	4	0.002	0.001	0.001	3.820
4	Medical Condition	6	0.001	0.001	0.001	1.458
5	Age	8	0.001	0.000	0.001	0.076
6	Test Results	3	0.001	0.000	0.000	1.733
7	Gender	2	0.000	0.000	0.000	0.978

What Went Well:

- I successfully prepared the dataset for analysis.
- I utilized the tool Orange effectively for my analysis.

What Did NOT Go Well:

- The algorithms I used yielded lower scores than expected
.
- There were limitations in the dataset that affected the analysis.

What Would I Do Differently Next Time:

- I will consider using different algorithms or adjusting parameters to improve scores.
- Next time, I will gather a more comprehensive dataset to ensure better analysis results.
- I will also ensure to implement formulas more effectively to enhance the accuracy of my analysis